Reflection and refraction – A contrast

9a

To contrast means to describe the differences between things. When we contrast information, it is generally helpful to use contrasting conjunctions such as *more than, less than, different from, in contrast to* and *rather.*

1 Can you think of other contrasting words? Write them on the line below.

Using these words or phrases in a sentence highlights differences. Differences between a concave and a convex mirror are indicated in each of the following sentences:

- A concave mirror curves inwards, while a convex mirror curves outwards.
- In contrast to concave mirrors, which curve inwards, convex mirrors curve outwards.

2 Write two contrasting sentences about mirrors using the information given in this table:

Feature	Concave mirror	Convex mirror
Mirror sketches		
Focus position	In front of mirror	Behind mirror
Image of close object	Upright, magnified	Upright, diminished
Image of distant object	Upside-down, diminished	Upright, diminished

3 Write two contrasting sentences about lenses using the information given in this table:

Feature	Concave lens	Convex lens
Sketch		
Image	Diminished	Magnified

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9b

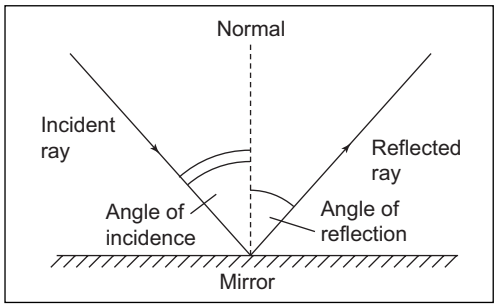
Reflection

Definition: Light that bounces off surfaces is reflected light.

Observation: Mirrors reflect light orderly and images are formed. Surfaces that are rough reflect light in all directions to giving overlapping images.

Law: The angle of reflection is equal to the angle of incidence.

Sketch:



Refraction

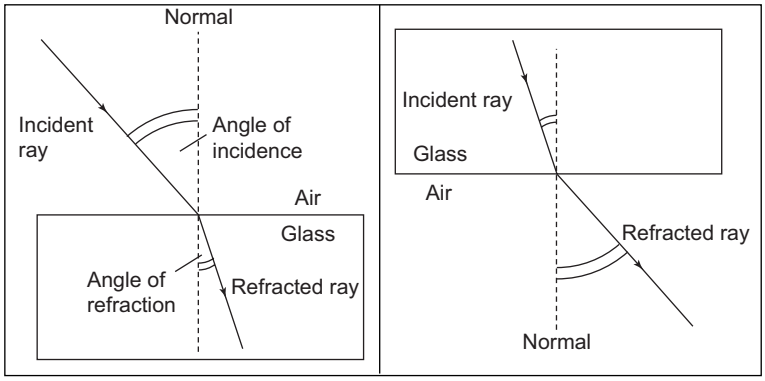
Definition: Light that bends as it enters another medium is refracted light

Observation: Smooth lenses make images look bigger or smaller (magnified with convex lenses, diminished with concave lenses).

Law:

- Light is bent towards the normal as it enters a denser medium, e.g. travels from air to glass.
- Light is bent away from the normal as it enters a less dense medium, e.g. as it passes from glass to air.

Sketches:



4 Look at the diagrams above and then define the following:

a Incident ray: _____

b Reflected ray: _____

c Refracted ray: _____

d Normal: _____

e Angle of incidence: _____

f Angle of reflection: _____

g Angle of refraction: _____

5 Look at the table about reflection and refraction, and then write five contrasting sentences.

_____

_____

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